BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

LESSON 143 · Lesson 143 · The TCP/IP Stack and Encapsulation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which device typically forwards traffic between different IP networks?

Q2. Which logic gate outputs 1 only when both inputs are 1?

Q3. What is the accumulator for?

Q4. Which is faster in the worst case on sorted arrays, linear or binary search?

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I Do · Read new knowledge

Key vocabulary

application layer — Protocols providing services to programs, such as HTTP or DNS.

transport layer — End-to-end service such as TCP or UDP.

internet layer — IP packet addressing and routing between networks.

link layer — Local frame delivery on physical or wireless link.

New learning

Application data — HTTP sends a request body/application message.

Transport and IP — TCP segments carry ports, IP packets carry network addresses.

Frame on local link — Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Worked example

A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

1. HTTP application data — Identify the relevant information and make a prediction.

2. TCP segment in IP packet — Explain the change using correct technical terms.

3. Ethernet frame — HTTP application bytes become TCP segment payload; the TCP segment goes into an IP packet; the packet is carried in an Ethernet frame. Receivers process headers in reverse to deliver data to the browser/server.

We Do · Practise with guidance

A packet passes through a router from Ethernet into a Wi-Fi network. Which information changes?

Incoming Ethernet frame

Identify the relevant information and make a prediction.

Route IP packet

Explain the change using correct technical terms.

Outgoing Wi-Fi frame

Router removes the incoming link-layer framing, processes/forwards the IP packet and builds new outgoing link-layer framing. IP hop limit/TTL normally decreases; end-to-end transport data need not change.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Name four layers of the TCP/IP stack and give one role for each. [8]

Question 2

Describe encapsulation of application data over TCP/IP and Ethernet. [6]

Question 3

Explain what a router changes when forwarding a packet to a different link technology. [8]

Exit Ticket · Answer before checking

Every router preserves the same link-layer frame all the way to the destination.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Router.

Q2. AND.

Q3. Holding a CPU's working data/ALU results.

Q4. Binary search: O(log n) rather than O(n).

You Do pages · Independent questions

Q1. Application user-facing network protocols [2], transport end-to-end delivery/ports [2], internet addressing/routing [2], link local frame transport [2].

Q2. Application message [1], TCP header/segment [2], IP header/packet [2], frame encapsulation [1].

Q3. Remove old link frame [2], inspect IP destination/routing [2], reduce TTL/hop limit and update necessary header fields [2], encapsulate with new link frame [2].

Exit Ticket · Model answer

False. Frames are recreated for each local link/hop.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: